

IMMUNE SYSTEM DISORDERS

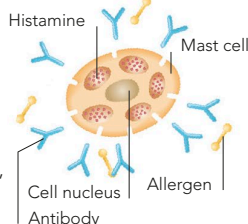
THE IMMUNE SYSTEM SOMETIMES OVER-REACTS, CAUSING AN ALLERGIC RESPONSE. WHEN THE SYSTEM IS WEAK, IMMUNIZATION CAN HELP TO BOOST IT. HOWEVER, IT MAY BECOME SO WEAK – BY AN HIV INFECTION, FOR EXAMPLE – THAT EVEN ORDINARY INFECTIONS CAN BE DANGEROUS.

ALLERGIES

When first exposed to an allergen, such as nuts or pollen, the immune system makes antibodies to fight it. The antibodies coat the surface of mast cells in the skin, stomach lining, lungs, and upper airways. If the allergen enters the body again, these cells mount an allergic response.

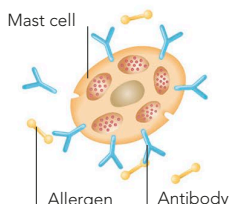
1 EXPOSURE

The first time an allergen enters the body, antibodies bind to the surface of mast cells. These cells contain histamine, which normally causes inflammation.



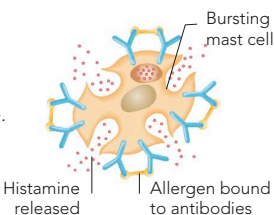
2 ANTIBODIES

When the allergen returns, the antibodies are triggered into action. If the allergen links two or more antibodies together, the cell bursts.



3 HISTAMINE

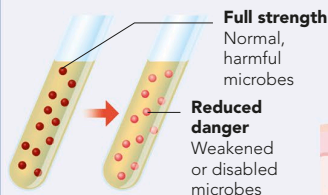
As the cell bursts, it releases histamine, which causes an inflammatory response. This irritates body tissues and produces all the symptoms of an allergy.



IMMUNIZATION

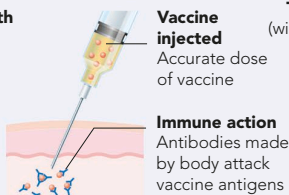
The process of becoming resistant or immune to a particular microbe as a result of infection is known as natural immunization. Resistance can also be developed artificially. In active immunization, dead or weakened versions of the microbe or its toxic products are injected into the body. The immune response occurs, with the production of antibodies,

but the illness does not develop. If urgent protection is needed, or if an immune system is weak, passive immunization can be used by injecting ready-made antibodies. These antibodies provide swift resistance against the microbes, but they gradually degenerate and are not replaced. The body has no memory for making them again.



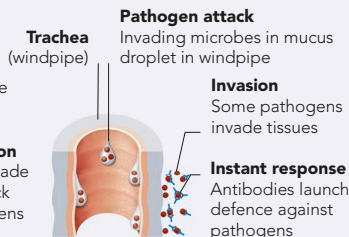
1 VACCINE PRODUCTION

A vaccine contains complete or partial microbes, or the toxins they make. It can stimulate the immune response but not cause symptoms.



2 VACCINE DELIVERY

Vaccination stimulates the body's immune system to raise antibodies against the antigens on the disease-carrying organisms.



3 IMMUNE RESPONSE

A pathogen against which the body has been vaccinated alerts the memory cells, and so the immune system launches an instant defence.